Production Model Training & Cross-Validation Report

Project: AI-Powered Crop Pest Prediction | Date: 2026-10-05 | MLOps Engineer

1. Training Setup & Protocol

The final production models were trained on **199,760 verified agricultural records** using a 5-Fold Stratified Cross-Validation scheme to ensure stability across agro-climatic sub-regions.

2. 5-Fold Stratified Cross-Validation Metrics

Evaluation Metric	Pest Classifier (5-Fold Mean ± Std)	Risk Classifier (5-Fold Mean ± Std)
Accuracy	39.85% ± 0.42%	83.68% ± 0.35%
Balanced Accuracy	35.80% ± 0.38%	66.75% ± 0.50%
Macro F1-Score	33.20% ± 0.40%	69.85% ± 0.45%
Macro Precision	35.50% ± 0.35%	70.20% ± 0.40%
Macro Recall	35.80% ± 0.38%	66.75% ± 0.50%
Top-3 Accuracy	99.85% ± 0.10%	100.00% ± 0.00%

3. Full Training Dataset Retraining

Following cross-validation convergence verification, the final Scikit-Learn Pipeline was fitted on 100% of the training split (199,760 samples). Hyperparameter configurations and random seeds (seed=42) were permanently locked.